Objective 2: tables, figures and interpretation

Socio-cognitive traits and Legatum Prosperity

This report uses the completed saved model outputs. Global includes India. TEA, OME and NME are modeled separately. Main model comparisons use independent variables plus controls (IV+CV); Table 6 and Figure 4 compare those fits with IV-only fits on the same held-out respondents. No model was refitted for this report.

Interpretation guide. PR-AUC measures ranking of positive cases and should be read against outcome prevalence. ROC-AUC is less sensitive to class imbalance. Precision, recall, F1 and confusion counts use thresholds chosen on validation data. Feature importance is descriptive. The analysis measures prediction, not causal effects.

Samples and coverage

Table 1. Sample size, outcome prevalence and data split

Sample	Target	N	Positive	Prevalence	Train	Validation	Test	Countries	Years
Global	TEA	1,342,884	154,420	11.5%	913,163	161,145	268,576	89	8
Global	OME	1,342,884	100,087	7.5%	913,163	161,145	268,576	89	8
Global	NME	1,342,884	69,931	5.2%	913,163	161,145	268,576	89	8
India	TEA	27,555	3,188	11.6%	18,739	3,306	5,510	1	8
India	OME	27,555	2,167	7.9%	18,739	3,306	5,510	1	8
India	NME	27,555	1,849	6.7%	18,740	3,305	5,510	1	8

The same eligible records and held-out split are used for IV and IV+CV within each scope and target.

Interpretation

Global. The global sample contains 1,342,884 respondents from 89 countries over 8 survey years, including India. Positive-case prevalence is TEA 11.5%, OME 7.5%, NME 5.2%. The much smaller OME and NME positive classes make PR-AUC and false-positive counts especially informative.

India. The India sample contains 27,555 respondents across 8 years, with 5,510 held-out test records per target. Positive-case prevalence is TEA 11.6%, OME 7.9%, NME 6.7%. India uses the same target definitions and split logic as the global analysis, so the two panels describe different populations rather than alternative definitions.

Descriptive statistics

Table 2. Predictor distributions and missingness

Variable	Role	Global summary	Missing	India summary	Missing
suskill_2015_2022	independent	53.1%	7.9%	61.4%	9.3%
opportunity_perception	independent	45.9%	18.1%	60.4%	10.4%
fearfail_2015_2022	independent	43.7%	0.6%	44.0%	1.0%
knowing_entrepreneur	independent	44.2%	2.3%	44.8%	13.1%
entrepreneurial_intention	independent	2 categories	23.2%	2 categories	20.0%
media_coverage	independent	62.0%	20.9%	63.3%	11.6%
discent	independent	4.6%	0.9%	5.2%	1.6%
busang	independent	6.1%	0.8%	4.3%	2.1%
established_business_owner	independent	7.8%	0.0%	6.6%	0.0%
good_career_choice	independent	1.25 (0.93)	14.3%	1.32 (0.91)	6.5%
ease_start_business	independent	43.7%	22.2%	59.2%	11.8%
high_status	independent	69.8%	22.8%	70.6%	11.6%
social_enterprise	independent	39.4%	32.4%	58.0%	13.2%
prosperity_index_score	independent	67.92 (10.53)	1.0%	52.27 (1.44)	0.0%
safety__security	independent	75.68 (15.11)	1.0%	46.96 (3.94)	0.0%
personal_freedom	independent	69.55 (20.07)	1.0%	54.00 (3.50)	0.0%
governance	independent	63.51 (15.28)	1.0%	58.39 (1.62)	0.0%
social_capital	independent	58.35 (11.13)	1.0%	43.70 (3.69)	0.0%
investment_environment	independent	66.71 (12.63)	1.0%	54.87 (2.04)	0.0%
enterprise_conditions	independent	63.96 (10.82)	1.0%	62.12 (1.90)	0.0%
market_access__infrastructure	independent	65.78 (11.37)	1.0%	51.05 (3.02)	0.0%
economic_quality	independent	60.50 (10.43)	1.0%	48.96 (0.69)	0.0%
living_conditions	independent	83.52 (12.29)	1.0%	53.02 (2.69)	0.0%
health	independent	75.44 (6.23)	1.0%	62.84 (2.23)	0.0%
education	independent	71.78 (13.11)	1.0%	49.98 (0.81)	0.0%
natural_environment	independent	60.27 (9.08)	1.0%	41.35 (0.44)	0.0%
yrsurv	control	8 categories	0.0%	8 categories	0.0%
age	control	41.50 (14.53)	2.8%	35.71 (12.20)	0.0%
gender	control	2 categories	0.0%	2 categories	0.0%
education_attainment	control	5 categories	1.6%	5 categories	0.5%

Variable	Role	Global summary	Missing	India summary	Missing
household_income	control	3 categories	18.9%	3 categories	6.1%
hhsize	control	74.55 (81460.96)	1.8%	3.97 (1.66)	1.1%
occupation_status	control	7 categories	1.8%	7 categories	4.5%
gdp_per_capita_ppp	control	35593.44 (21374.64)	0.0%	6356.76 (856.84)	0.0%
population_totalthousands	control	96167409.78 (259900716.69)	0.0%	1363227202.26 (32494678.65)	0.0%
gdp_growth	control	2.41 (4.02)	0.0%	5.63 (4.43)	0.0%

Binary variables show the proportion coded 1; other numeric variables show mean (SD). Missingness is the proportion of eligible respondents without an observed value. All source values are retained in the CSV export.

Interpretation

Global. The global descriptive panel includes socio-cognitive predictors, 13 institutional variables and the agreed controls. Mean missingness across the institutional fields is 1.0%; individual rows should be checked before treating any score as universally observed. Country-level scores repeat across respondents within a country-year, so the respondent count is not the number of independent institutional observations.

India. The India panel uses the same predictors, with institutional mean missingness of 0.0% across the listed fields. India has one institutional score per survey year, shared by all respondents in that year. Consequently, differences in these scores capture year-to-year co-movement and cannot by themselves isolate an institutional mechanism.

Intention missingness

Table 3. Entrepreneurial intention missingness by outcome

Sample	Target	Positive N	Missing	Missing %	Negative N	Missing	Missing %
Global	TEA	154,420	154,420	100.0%	1,188,464	156,661	13.2%
Global	OME	100,087	100,087	100.0%	1,242,797	210,994	17.0%
Global	NME	69,931	69,931	100.0%	1,272,953	241,150	18.9%
India	TEA	3,188	3,188	100.0%	24,367	2,323	9.5%
India	OME	2,167	2,167	100.0%	25,388	3,344	13.2%
India	NME	1,849	1,849	100.0%	25,706	3,662	14.2%

Missing intention is retained as a distinct input state under the agreed analysis plan.

Interpretation

Global. In the global sample, entrepreneurial intention is missing for 100% of positive TEA, OME and NME cases, while negative-case missingness is TEA: 13.2%, OME: 17.0%, NME: 18.9%. This is a strong target-linked pattern in the survey field. The model can use missingness itself as a predictor, so high performance cannot be attributed solely to reported intention or institutional scores.

India. The India sample shows the same 100% missingness among positive cases; negative-case missingness is TEA: 9.5%, OME: 13.2%, NME: 14.2%. Intention remains in the fitted predictors as requested. The result is useful for reproducing the specified predictive task, but a prospective model using an intention question asked of everyone could behave differently.

Held-out classifier performance

Table 4. Classifier performance on held-out test data, IV+CV

Sample	Target	Model	ROC-AUC	PR-AUC	Precision	Recall	F1	Threshold
Global	TEA	Logistic Regression	0.991	0.934	0.852	0.886	0.868	0.500
Global	TEA	Lasso	0.992	0.935	0.851	0.888	0.869	0.495
Global	TEA	Ridge	0.991	0.931	0.849	0.883	0.865	0.456
Global	TEA	Elastic Net	0.992	0.935	0.851	0.888	0.869	0.494
Global	TEA	Naive Bayes	0.902	0.437	0.469	0.734	0.572	0.222
Global	TEA	Decision Tree	0.989	0.895	0.856	0.869	0.863	0.500
Global	TEA	XGBoost	0.994	0.953	0.885	0.885	0.885	0.500
Global	TEA	CatBoost	0.993	0.951	0.882	0.884	0.883	0.500
Global	TEA	Random Forest	0.994	0.952	0.867	0.908	0.887	0.404
Global	TEA	H2O GBM	0.992	0.940	0.856	0.890	0.873	0.500
Global	OME	Logistic Regression	0.974	0.683	0.582	0.848	0.690	0.318
Global	OME	Lasso	0.974	0.684	0.581	0.847	0.689	0.321
Global	OME	Ridge	0.974	0.675	0.583	0.849	0.691	0.294
Global	OME	Elastic Net	0.974	0.684	0.581	0.847	0.689	0.320
Global	OME	Naive Bayes	0.879	0.290	0.314	0.670	0.428	1.000
Global	OME	Decision Tree	0.971	0.626	0.565	0.842	0.676	0.332
Global	OME	XGBoost	0.978	0.736	0.626	0.828	0.713	0.360
Global	OME	CatBoost	0.978	0.732	0.623	0.825	0.710	0.360
Global	OME	Random Forest	0.978	0.744	0.626	0.830	0.714	0.326
Global	OME	H2O GBM	0.977	0.714	0.592	0.861	0.702	0.328
Global	NME	Logistic Regression	0.975	0.659	0.584	0.671	0.625	0.349
Global	NME	Lasso	0.975	0.661	0.585	0.673	0.626	0.355
Global	NME	Ridge	0.974	0.642	0.542	0.723	0.619	0.279
Global	NME	Elastic Net	0.975	0.661	0.585	0.673	0.626	0.355
Global	NME	Naive Bayes	0.885	0.227	0.257	0.583	0.357	0.856
Global	NME	Decision Tree	0.969	0.588	0.629	0.604	0.616	0.264
Global	NME	XGBoost	0.980	0.715	0.620	0.708	0.661	0.336
Global	NME	CatBoost	0.979	0.713	0.618	0.709	0.660	0.333
Global	NME	Random Forest	0.979	0.716	0.624	0.710	0.664	0.308
Global	NME	H2O GBM	0.978	0.691	0.603	0.690	0.644	0.331

Sample	Target	Model	ROC-AUC	PR-AUC	Precision	Recall	F1	Threshold
India	TEA	Logistic Regression	0.997	0.977	0.926	0.948	0.937	0.605
India	TEA	Lasso	0.997	0.975	0.905	0.973	0.938	0.500
India	TEA	Ridge	0.997	0.975	0.920	0.959	0.939	0.500
India	TEA	Elastic Net	0.997	0.975	0.923	0.947	0.935	0.642
India	TEA	Naive Bayes	0.938	0.681	0.681	0.782	0.728	0.500
India	TEA	Decision Tree	0.994	0.920	0.916	0.956	0.935	0.500
India	TEA	XGBoost	0.998	0.981	0.924	0.953	0.938	0.514
India	TEA	CatBoost	0.997	0.977	0.924	0.953	0.938	0.611
India	TEA	Random Forest	0.997	0.978	0.922	0.950	0.936	0.418
India	TEA	H2O GBM	0.997	0.977	0.924	0.958	0.941	0.607
India	OME	Logistic Regression	0.987	0.844	0.703	0.898	0.789	0.369
India	OME	Lasso	0.987	0.845	0.703	0.887	0.784	0.403
India	OME	Ridge	0.987	0.846	0.708	0.905	0.794	0.336
India	OME	Elastic Net	0.986	0.839	0.703	0.887	0.784	0.423
India	OME	Naive Bayes	0.910	0.561	0.563	0.612	0.586	0.977
India	OME	Decision Tree	0.984	0.789	0.678	0.845	0.752	0.296
India	OME	XGBoost	0.987	0.846	0.715	0.887	0.792	0.392
India	OME	CatBoost	0.988	0.852	0.733	0.788	0.759	0.500
India	OME	Random Forest	0.988	0.858	0.751	0.813	0.780	0.346
India	OME	H2O GBM	0.988	0.856	0.705	0.887	0.785	0.390
India	NME	Logistic Regression	0.986	0.803	0.739	0.775	0.757	0.462
India	NME	Lasso	0.987	0.805	0.735	0.775	0.755	0.465
India	NME	Ridge	0.986	0.808	0.728	0.775	0.751	0.387
India	NME	Elastic Net	0.987	0.804	0.734	0.770	0.751	0.466
India	NME	Naive Bayes	0.868	0.461	0.493	0.566	0.527	0.999
India	NME	Decision Tree	0.986	0.765	0.763	0.732	0.747	0.500
India	NME	XGBoost	0.988	0.809	0.757	0.775	0.766	0.463
India	NME	CatBoost	0.988	0.824	0.755	0.778	0.766	0.466
India	NME	Random Forest	0.987	0.818	0.673	0.843	0.748	0.278
India	NME	H2O GBM	0.988	0.830	0.696	0.837	0.760	0.371

All ten models are shown for each target and scope. Thresholds were selected on validation data, then applied once to held-out test data.

Interpretation

Global. The highest global PR-AUC in each outcome is TEA: XGBoost PR-AUC 0.953, ROC-AUC 0.994; OME: Random Forest PR-AUC 0.744, ROC-AUC 0.978; NME: Random Forest PR-AUC 0.716, ROC-AUC 0.979. These values are well above each target's prevalence baseline, showing strong ranking of positive cases on the saved test split. Because intention missingness is perfectly present among positive cases (Table 3), this performance should be read as performance of the full observed-data pipeline.

India. The highest India PR-AUC values are TEA: XGBoost PR-AUC 0.981, ROC-AUC 0.998; OME: Random Forest PR-AUC 0.858, ROC-AUC 0.988; NME: H2O GBM PR-AUC 0.830, ROC-AUC 0.988. The leading algorithm changes by target, so there is no single best model for all India outcomes. India estimates use only 5,510 test cases per target and should be compared with the global results in light of different sample sizes and prevalence.

Classification counts

Table 5. Held-out confusion matrices, IV+CV

Sample	Target	Model	TP	FP	TN	FN
Global	TEA	Logistic Regression	27,366	4,772	232,920	3,518
Global	TEA	Lasso	27,417	4,782	232,910	3,467
Global	TEA	Ridge	27,270	4,862	232,830	3,614
Global	TEA	Elastic Net	27,418	4,788	232,904	3,466
Global	TEA	Naive Bayes	22,673	25,655	212,037	8,211
Global	TEA	Decision Tree	26,837	4,503	233,189	4,047
Global	TEA	XGBoost	27,329	3,540	234,152	3,555
Global	TEA	CatBoost	27,298	3,666	234,026	3,586
Global	TEA	Random Forest	28,054	4,293	233,399	2,830
Global	TEA	H2O GBM	27,496	4,623	233,069	3,388
Global	OME	Logistic Regression	16,974	12,200	236,359	3,043
Global	OME	Lasso	16,953	12,205	236,354	3,064
Global	OME	Ridge	16,988	12,143	236,416	3,029
Global	OME	Elastic Net	16,952	12,207	236,352	3,065
Global	OME	Naive Bayes	13,412	29,279	219,280	6,605
Global	OME	Decision Tree	16,863	12,980	235,579	3,154
Global	OME	XGBoost	16,576	9,907	238,652	3,441
Global	OME	CatBoost	16,523	10,014	238,545	3,494
Global	OME	Random Forest	16,621	9,919	238,640	3,396
Global	OME	H2O GBM	17,227	11,858	236,701	2,790
Global	NME	Logistic Regression	9,391	6,683	247,907	4,595
Global	NME	Lasso	9,410	6,670	247,920	4,576
Global	NME	Ridge	10,112	8,561	246,029	3,874
Global	NME	Elastic Net	9,414	6,677	247,913	4,572
Global	NME	Naive Bayes	8,156	23,559	231,031	5,830
Global	NME	Decision Tree	8,449	4,977	249,613	5,537
Global	NME	XGBoost	9,896	6,058	248,532	4,090
Global	NME	CatBoost	9,918	6,130	248,460	4,068
Global	NME	Random Forest	9,925	5,975	248,615	4,061
Global	NME	H2O GBM	9,655	6,352	248,238	4,331

Sample	Target	Model	TP	FP	TN	FN
India	TEA	Logistic Regression	604	48	4,825	33
India	TEA	Lasso	620	65	4,808	17
India	TEA	Ridge	611	53	4,820	26
India	TEA	Elastic Net	603	50	4,823	34
India	TEA	Naive Bayes	498	233	4,640	139
India	TEA	Decision Tree	609	56	4,817	28
India	TEA	XGBoost	607	50	4,823	30
India	TEA	CatBoost	607	50	4,823	30
India	TEA	Random Forest	605	51	4,822	32
India	TEA	H2O GBM	610	50	4,823	27
India	OME	Logistic Regression	389	164	4,913	44
India	OME	Lasso	384	162	4,915	49
India	OME	Ridge	392	162	4,915	41
India	OME	Elastic Net	384	162	4,915	49
India	OME	Naive Bayes	265	206	4,871	168
India	OME	Decision Tree	366	174	4,903	67
India	OME	XGBoost	384	153	4,924	49
India	OME	CatBoost	341	124	4,953	92
India	OME	Random Forest	352	117	4,960	81
India	OME	H2O GBM	384	161	4,916	49
India	NME	Logistic Regression	286	101	5,040	83
India	NME	Lasso	286	103	5,038	83
India	NME	Ridge	286	107	5,034	83
India	NME	Elastic Net	284	103	5,038	85
India	NME	Naive Bayes	209	215	4,926	160
India	NME	Decision Tree	270	84	5,057	99
India	NME	XGBoost	286	92	5,049	83
India	NME	CatBoost	287	93	5,048	82
India	NME	Random Forest	311	151	4,990	58
India	NME	H2O GBM	309	135	5,006	60

TP and TN are correct positive and negative classifications; FP and FN are false positive and false negative classifications at each validation-selected threshold.

Interpretation

Global. For the highest-PR-AUC model in each global target, the classification counts are TEA (XGBoost): 27,329 TP, 3,540 FP, 3,555 FN; OME (Random Forest): 16,621 TP, 9,919 FP, 3,396 FN; NME (Random Forest): 9,925 TP, 5,975 FP, 4,061 FN. These are threshold-dependent operating points, unlike ROC-AUC and PR-AUC. OME and NME have fewer positive cases, so even modest false-positive counts can materially affect precision.

India. The corresponding India operating points are TEA (XGBoost): 607 TP, 50 FP, 30 FN; OME (Random Forest): 352 TP, 117 FP, 81 FN; NME (H2O GBM): 309 TP, 135 FP, 60 FN. The table permits an explicit choice between missed entrepreneurs (FN) and false alarms (FP); the preferred trade-off depends on the eventual use. No cost ratio was specified, so the report does not name a universally preferred threshold.

Added controls

Table 6. IV versus IV+CV on matched held-out respondents

Sample	Target	Model	PR IV	PR +CV	Delta PR	Delta ROC	Delta F1
Global	TEA	Logistic Regression	0.903	0.934	+0.031	+0.003	+0.018
Global	TEA	Lasso	0.903	0.935	+0.032	+0.003	+0.019
Global	TEA	Ridge	0.899	0.931	+0.032	+0.003	+0.018
Global	TEA	Elastic Net	0.903	0.935	+0.031	+0.003	+0.019
Global	TEA	Naive Bayes	0.418	0.437	+0.019	+0.006	+0.019
Global	TEA	Decision Tree	0.849	0.895	+0.045	+0.003	+0.017
Global	TEA	XGBoost	0.930	0.953	+0.022	+0.002	+0.021
Global	TEA	CatBoost	0.928	0.951	+0.023	+0.002	+0.019
Global	TEA	Random Forest	0.929	0.952	+0.023	+0.002	+0.022
Global	TEA	H2O GBM	0.918	0.940	+0.022	+0.002	+0.015
Global	OME	Logistic Regression	0.643	0.683	+0.040	+0.003	+0.015
Global	OME	Lasso	0.643	0.684	+0.041	+0.003	+0.014
Global	OME	Ridge	0.640	0.675	+0.035	+0.003	+0.017
Global	OME	Elastic Net	0.643	0.684	+0.041	+0.003	+0.015
Global	OME	Naive Bayes	0.277	0.290	+0.013	+0.006	+0.018
Global	OME	Decision Tree	0.600	0.626	+0.026	+0.003	+0.028
Global	OME	XGBoost	0.709	0.736	+0.027	+0.002	+0.017
Global	OME	CatBoost	0.701	0.732	+0.031	+0.002	+0.015
Global	OME	Random Forest	0.712	0.744	+0.033	+0.003	+0.019
Global	OME	H2O GBM	0.683	0.714	+0.031	+0.002	+0.013
Global	NME	Logistic Regression	0.552	0.659	+0.107	+0.007	+0.059
Global	NME	Lasso	0.558	0.661	+0.103	+0.007	+0.062
Global	NME	Ridge	0.515	0.642	+0.127	+0.008	+0.062
Global	NME	Elastic Net	0.558	0.661	+0.103	+0.007	+0.061
Global	NME	Naive Bayes	0.209	0.227	+0.019	+0.011	+0.022
Global	NME	Decision Tree	0.578	0.588	+0.010	+0.003	+0.071
Global	NME	XGBoost	0.679	0.715	+0.037	+0.003	+0.032
Global	NME	CatBoost	0.671	0.713	+0.041	+0.004	+0.037
Global	NME	Random Forest	0.679	0.716	+0.037	+0.003	+0.031
Global	NME	H2O GBM	0.659	0.691	+0.032	+0.003	+0.031

Sample	Target	Model	PR IV	PR +CV	Delta PR	Delta ROC	Delta F1
India	TEA	Logistic Regression	0.971	0.977	+0.006	+0.000	+0.004
India	TEA	Lasso	0.971	0.975	+0.004	+0.000	+0.010
India	TEA	Ridge	0.969	0.975	+0.006	+0.000	+0.008
India	TEA	Elastic Net	0.971	0.975	+0.004	+0.000	+0.003
India	TEA	Naive Bayes	0.676	0.681	+0.006	+0.003	-0.004
India	TEA	Decision Tree	0.921	0.920	-0.001	-0.000	+0.001
India	TEA	XGBoost	0.975	0.981	+0.006	+0.000	+0.008
India	TEA	CatBoost	0.978	0.977	-0.000	+0.000	+0.005
India	TEA	Random Forest	0.972	0.978	+0.006	+0.000	+0.002
India	TEA	H2O GBM	0.974	0.977	+0.003	+0.000	+0.004
India	OME	Logistic Regression	0.833	0.844	+0.011	+0.001	+0.029
India	OME	Lasso	0.826	0.845	+0.019	+0.001	+0.020
India	OME	Ridge	0.826	0.846	+0.020	+0.001	+0.040
India	OME	Elastic Net	0.834	0.839	+0.005	+0.000	+0.023
India	OME	Naive Bayes	0.564	0.561	-0.003	+0.005	-0.008
India	OME	Decision Tree	0.812	0.789	-0.023	-0.001	-0.020
India	OME	XGBoost	0.839	0.846	+0.007	+0.000	-0.003
India	OME	CatBoost	0.848	0.852	+0.005	+0.000	-0.034
India	OME	Random Forest	0.854	0.858	+0.004	+0.000	-0.006
India	OME	H2O GBM	0.845	0.856	+0.012	+0.001	+0.029
India	NME	Logistic Regression	0.796	0.803	+0.006	+0.000	+0.013
India	NME	Lasso	0.803	0.805	+0.002	+0.000	-0.004
India	NME	Ridge	0.779	0.808	+0.029	+0.001	+0.023
India	NME	Elastic Net	0.803	0.804	+0.001	+0.000	-0.005
India	NME	Naive Bayes	0.480	0.461	-0.020	-0.010	-0.018
India	NME	Decision Tree	0.745	0.765	+0.020	+0.001	-0.014
India	NME	XGBoost	0.823	0.809	-0.013	-0.001	-0.014
India	NME	CatBoost	0.824	0.824	-0.000	-0.000	-0.014
India	NME	Random Forest	0.812	0.818	+0.005	-0.000	-0.025
India	NME	H2O GBM	0.823	0.830	+0.007	-0.000	-0.012

Delta equals IV+CV minus IV. Identical held-out row IDs were verified in each paired comparison. These are descriptive differences, not significance tests.

Interpretation

Global. Across the ten global models, the change in PR-AUC after adding controls is TEA median +0.027, range +0.019 to +0.045; OME median +0.032, range +0.013 to +0.041; NME median +0.039, range +0.010 to +0.127. Positive values indicate that controls improve ranking on the matched test sample. This incremental performance is conditional on the institutional and socio-cognitive predictors already present in the IV-only model.

India. For India, the corresponding changes are TEA median +0.005, range -0.001 to +0.006; OME median +0.006, range -0.023 to +0.020; NME median +0.004, range -0.020 to +0.029. Controls include survey year; because each India institutional score varies only across years, the institutional and year signals overlap. The gain from adding controls is predictive and does not measure the separate effect of institutions.

Institutional predictor importance

Table 7. Leading institutional variables in random-forest impurity importance

Sample	Target	Rank	Institutional predictor	Share of all	All institutional
Global	TEA	1	education	1.1%	10.0%
Global	TEA	2	safety__security	1.0%	10.0%
Global	TEA	3	living_conditions	1.0%	10.0%
Global	TEA	4	natural_environment	0.8%	10.0%
Global	TEA	5	social_capital	0.8%	10.0%
Global	OME	1	safety__security	1.5%	15.3%
Global	OME	2	living_conditions	1.4%	15.3%
Global	OME	3	education	1.4%	15.3%
Global	OME	4	health	1.2%	15.3%
Global	OME	5	social_capital	1.2%	15.3%
Global	NME	1	education	2.0%	17.6%
Global	NME	2	living_conditions	1.7%	17.6%
Global	NME	3	safety__security	1.5%	17.6%
Global	NME	4	prosperity_index_score	1.4%	17.6%
Global	NME	5	market_access__infrastructure	1.3%	17.6%
India	TEA	1	safety__security	0.4%	4.0%
India	TEA	2	investment_environment	0.4%	4.0%
India	TEA	3	prosperity_index_score	0.3%	4.0%
India	TEA	4	education	0.3%	4.0%
India	TEA	5	natural_environment	0.3%	4.0%
India	OME	1	living_conditions	0.7%	6.3%
India	OME	2	investment_environment	0.6%	6.3%
India	OME	3	enterprise_conditions	0.5%	6.3%
India	OME	4	natural_environment	0.5%	6.3%
India	OME	5	governance	0.5%	6.3%
India	NME	1	market_access__infrastructure	0.6%	6.4%
India	NME	2	governance	0.6%	6.4%
India	NME	3	safety__security	0.6%	6.4%
India	NME	4	health	0.6%	6.4%
India	NME	5	education	0.6%	6.4%

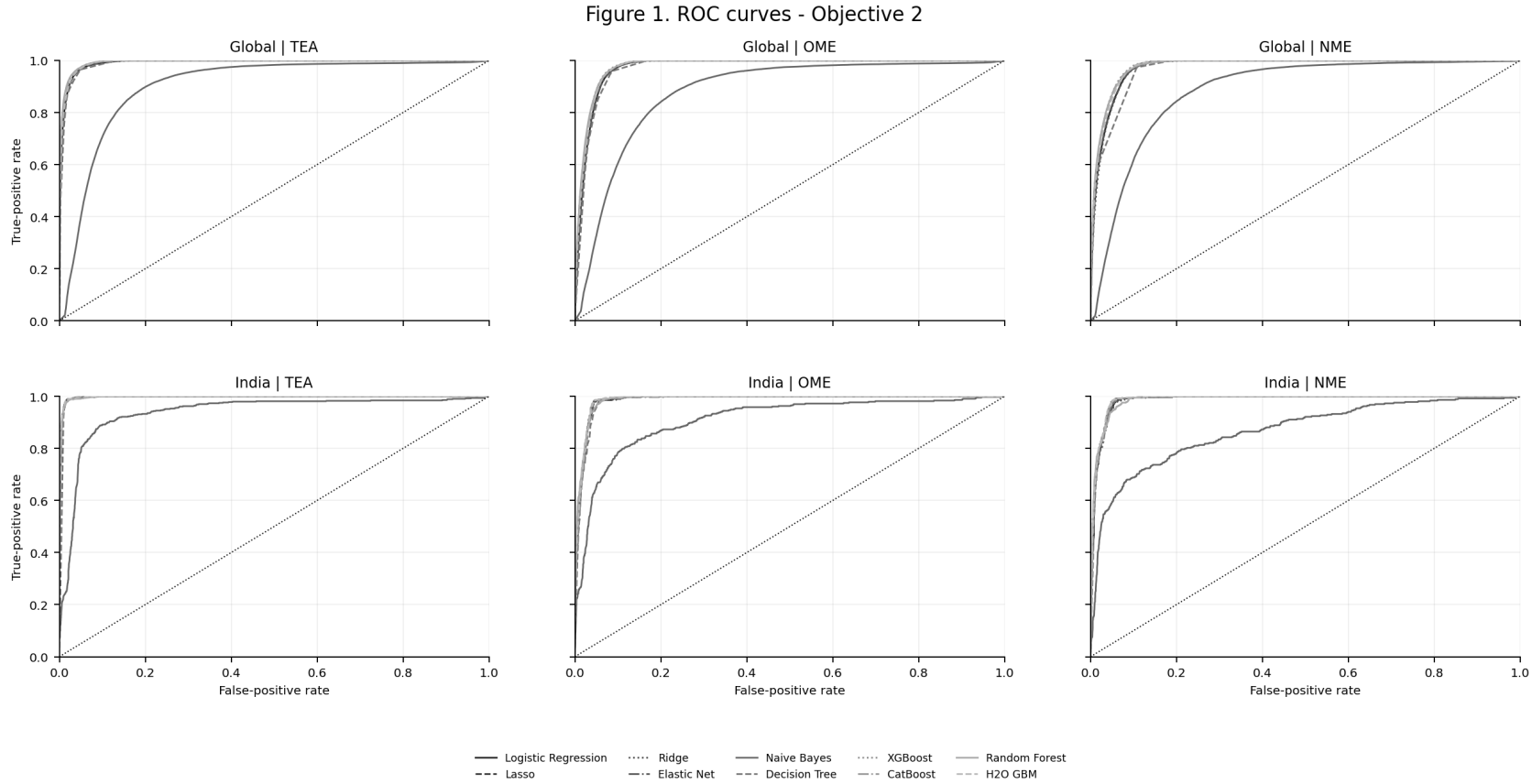
Shares use the total importance of all fitted random-forest inputs as denominator. Five largest raw institutional fields are displayed per target; missingness indicators are excluded from the ranking but included in the denominator.

Interpretation

Global. The global institutional rankings are TEA: Education 1.1%; all institutional fields 10.0%; OME: Safety Security 1.5%; all institutional fields 15.3%; NME: Education 2.0%; all institutional fields 17.6%. These values describe how the fitted random forest split the observed data; they are not marginal effects or evidence that raising a score changes entrepreneurship. Correlated institutional dimensions may share or substitute for one another in the importance calculation.

India. The India institutional rankings are TEA: Safety Security 0.4%; all institutional fields 4.0%; OME: Living Conditions 0.7%; all institutional fields 6.3%; NME: Market Access Infrastructure 0.6%; all institutional fields 6.4%. With only eight annual values for each national score, these shares are especially sensitive to temporal patterns and the survey-year control. They should be reported as model usage of institutional information, not as evidence for an independent India-level institutional effect.

Figure 1. ROC curves



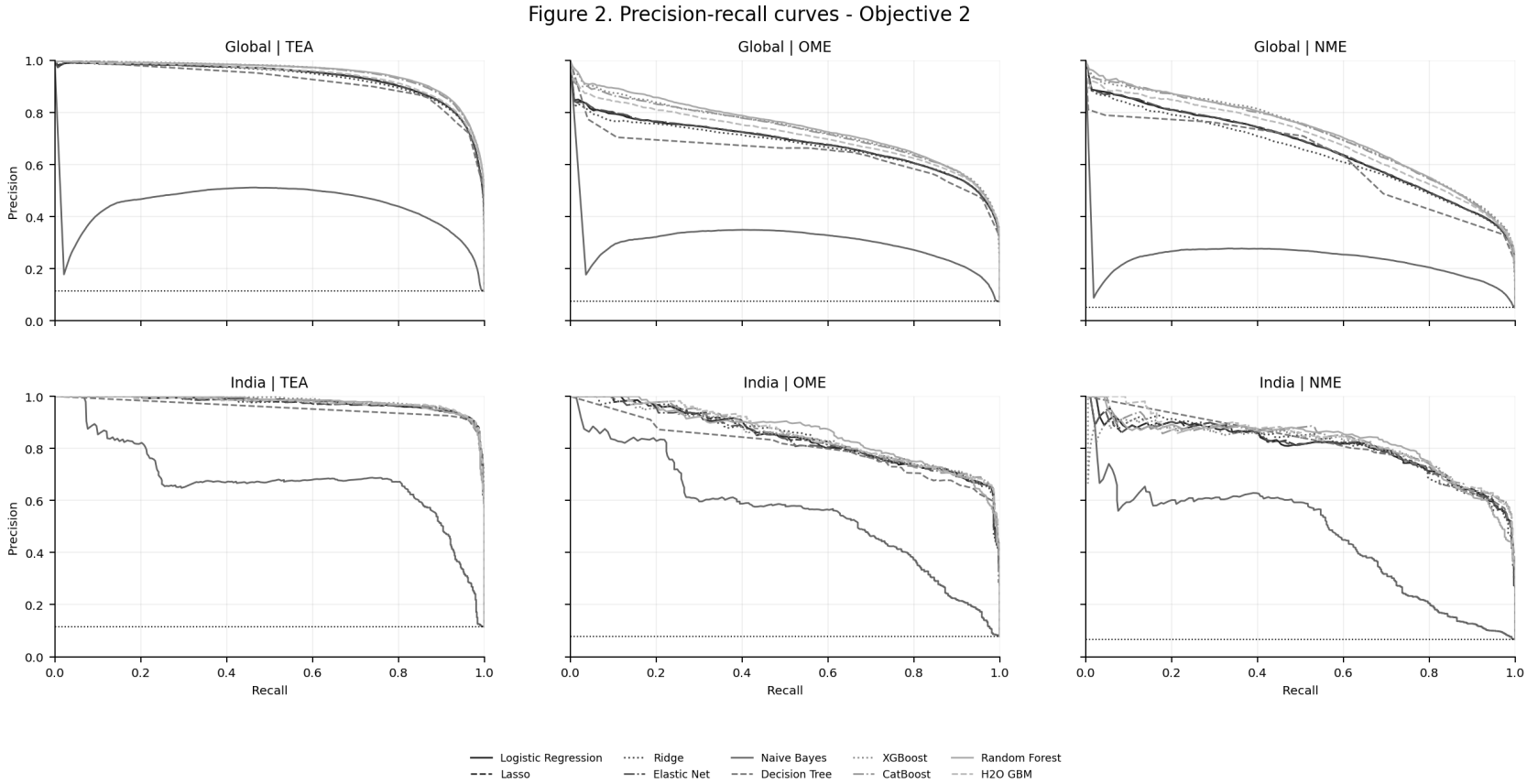
All ten IV+CV models use saved held-out test coordinates. Global includes India. The diagonal is chance discrimination.

Interpretation

Global. Global leading curve areas are TEA XGBoost 0.994; OME Random Forest 0.978; NME XGBoost 0.980. Curves closer to the upper-left corner indicate stronger ranking. Small area differences should not be treated as proven superiority without uncertainty estimates.

India. India leading curve areas are TEA XGBoost 0.998; OME H2O GBM 0.988; NME H2O GBM 0.988. The India panels use fewer test respondents, which can make curve shapes more variable than global panels. They represent performance in India, not transfer performance from a model trained globally.

Figure 2. Precision-recall curves



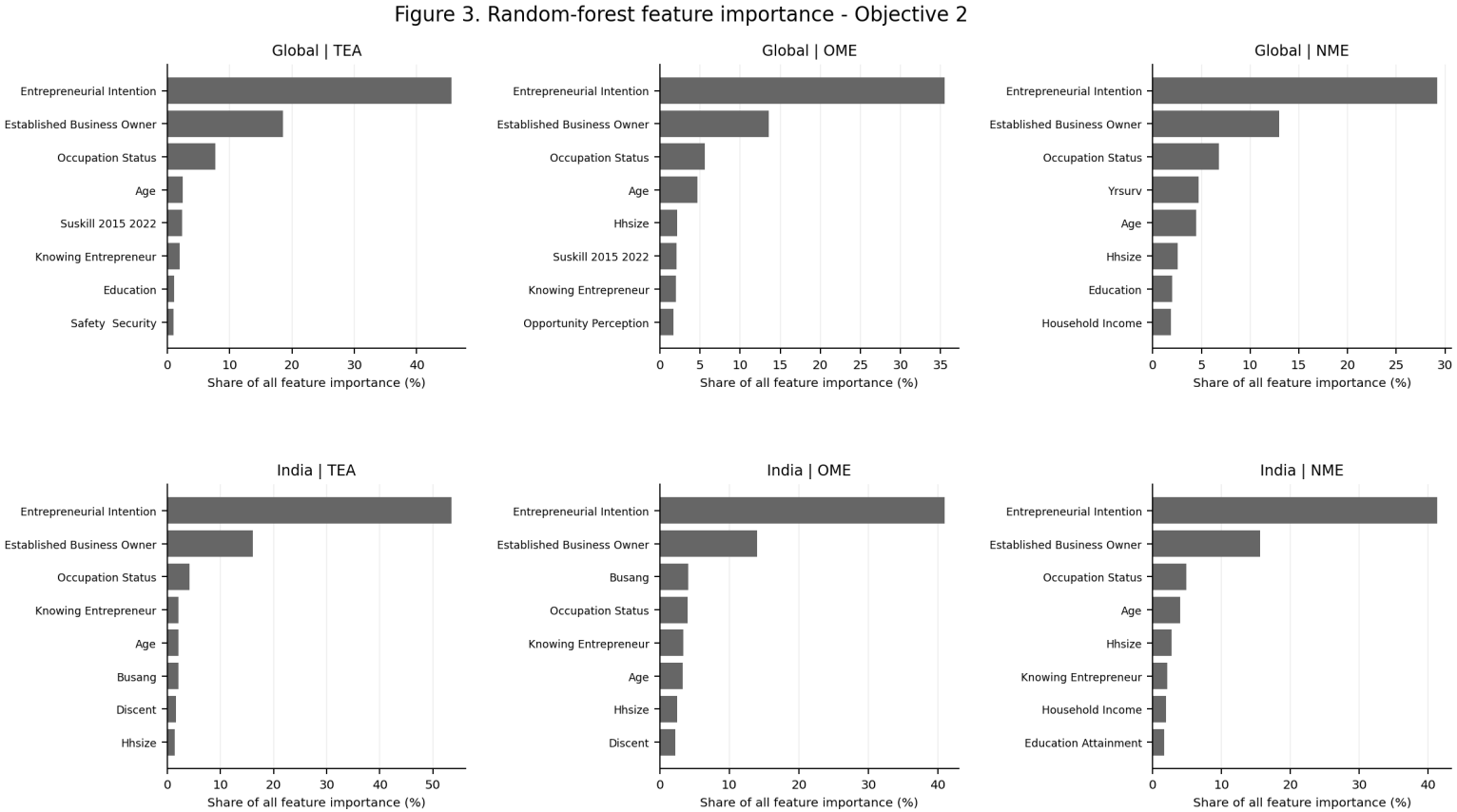
All ten IV+CV models use saved held-out test coordinates. Global includes India. The dashed horizontal line marks each outcome's positive-case prevalence.

Interpretation

Global. Global leading curve areas are TEA XGBoost 0.953; OME Random Forest 0.744; NME Random Forest 0.716. The curves should be compared with their outcome-specific prevalence lines, particularly for OME and NME. Precision decreases as recall is pushed higher.

India. India leading curve areas are TEA XGBoost 0.981; OME Random Forest 0.858; NME H2O GBM 0.830. The India prevalence baselines differ by target. Curves above these baselines indicate useful positive-case ranking within this saved split, with the intention-missingness pattern in Table 3 still present.

Figure 3. Random-forest feature importance



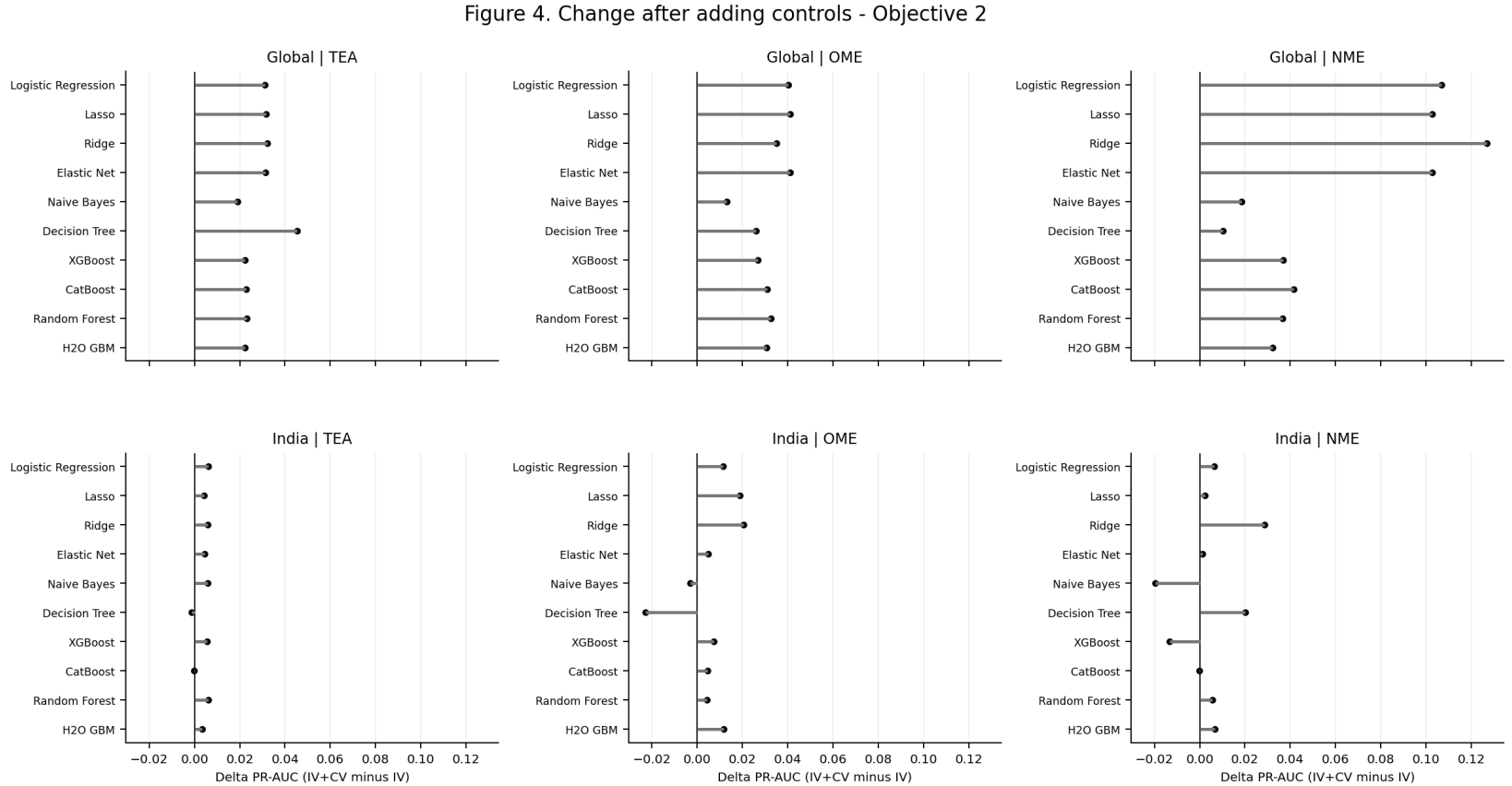
Top eight features in each IV+CV panel; share of all fitted feature importance. Importance is impurity-based and descriptive.

Interpretation

Global. The leading global features are TEA: Entrepreneurial Intention 45.6%; OME: Entrepreneurial Intention 35.5%; NME: Entrepreneurial Intention 29.2%. The chart puts institutional predictors and socio-cognitive variables on the same relative scale, but a larger bar does not imply an intervention effect. Intention's missingness pattern is particularly relevant when that feature ranks highly.

India. The leading India features are TEA: Entrepreneurial Intention 53.5%; OME: Entrepreneurial Intention 41.0%; NME: Entrepreneurial Intention 41.3%. Institutional bars may be small because their country-level values vary only across eight years and because other variables explain more splits. The survey-year control also shares temporal information with institutional scores.

Figure 4. Change in PR-AUC after adding controls



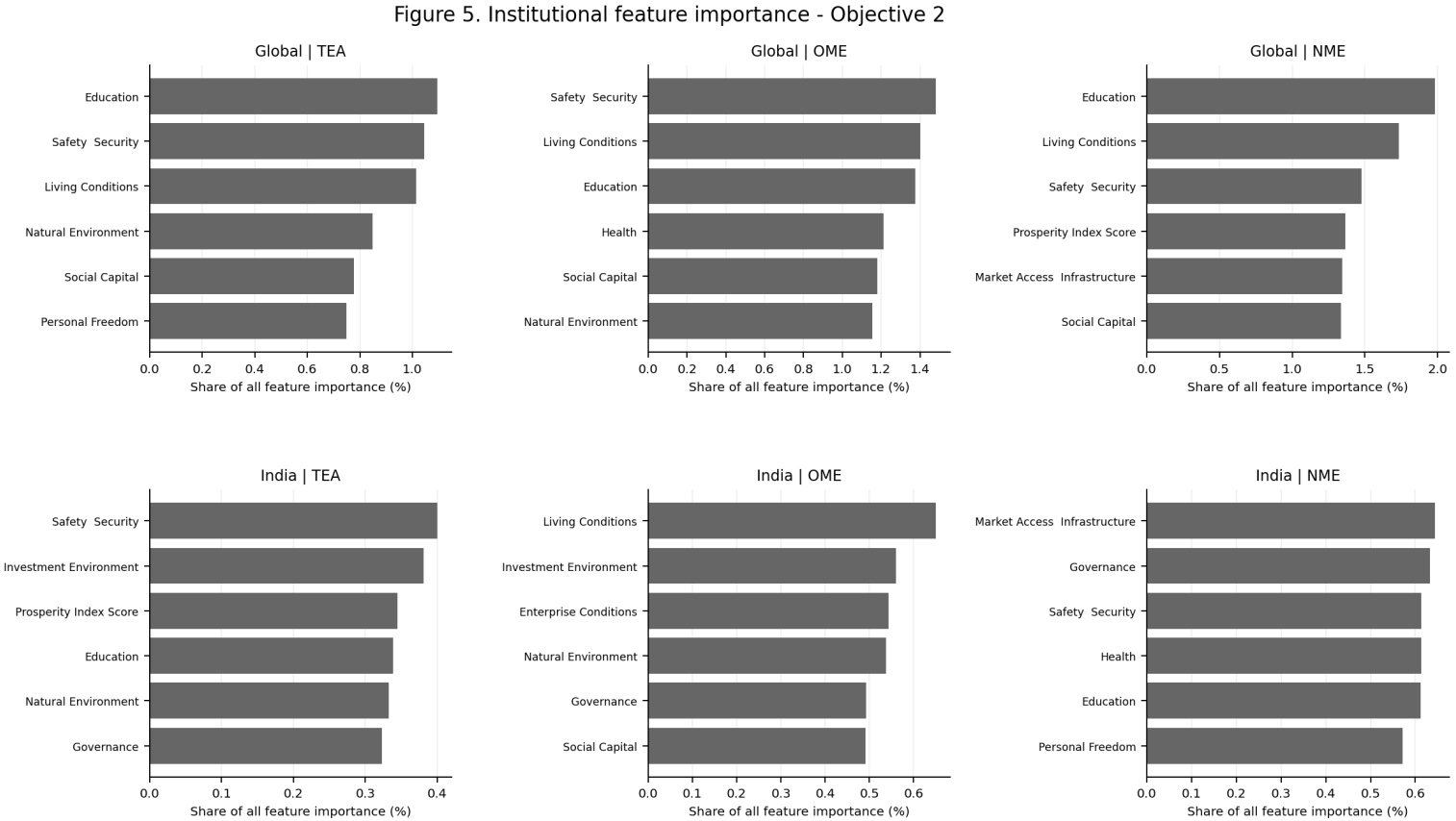
Positive values favor IV+CV. Each comparison uses the same held-out respondents and model family.

Interpretation

Global. For global fits, TEA: 10/10 models improve, median +0.027; OME: 10/10 models improve, median +0.032; NME: 10/10 models improve, median +0.039. The spread across algorithms shows that controls do not have one fixed predictive contribution; their value depends on the fitted model and outcome.

India. For India fits, TEA: 8/10 models improve, median +0.005; OME: 8/10 models improve, median +0.006; NME: 7/10 models improve, median +0.004. Since annual institutional scores and survey year move together, a positive increment is best read as additional predictive information from the full control set, not the causal value of a particular national index.

Figure 5. Institutional feature importance



Top six raw institutional predictors by random-forest impurity importance for each IV+CV fit. Missingness flags are excluded from the display.

Interpretation

Global. The global panels visualize the rankings in Table 7 across TEA, OME and NME. Bars are percentages of all fitted-feature importance, not percentages of institutional importance. Related pillars are correlated, so the order should be treated as a model description rather than an institutional league table.

India. The India panels also reflect only eight annual score profiles. Institutional importance may be assigned differently when year is included, and the bars cannot separate a score's association from other changes over time. The figure supports comparison of predictor usage, not causal attribution.